

Calcolo Scientifico, esercizi su sistemi lineari

Risolvere i seguenti sistemi lineari:

$$1. A = \begin{pmatrix} 6 & 2 & 0 \\ 2 & -12 & 3 \\ 1 & -2 & 4 \end{pmatrix}, b = \begin{pmatrix} -31 \\ 0 \\ 13 \end{pmatrix}, sol = \begin{pmatrix} -\frac{690}{131} \\ \frac{79}{262} \\ \frac{618}{131} \end{pmatrix};$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive	2.1078e-17	-262	-5.267175572519084 0.301526717557252 4.717557251908397
Gauss-Piv	2.1078e-17	-262	-5.267175572519084 0.301526717557252 4.717557251908397

Indice di condizionamento: 4.046231122814087

$$2. A = \begin{pmatrix} 9 & 2 & 1.2 & -1.88 \\ 3.4 & 5.6 & 0 & 0 \\ -4.2 & 1 & 14.5 & 2 \\ 0 & 2 & 3 & 15 \end{pmatrix}, b = \begin{pmatrix} 9.08 \\ 14.6 \\ 49.3 \\ 73 \end{pmatrix}, sol = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix};$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive	2.2204e-16	9.656100799999998e+03	1.0000000000000000 2.0000000000000000 3.0000000000000000 4.0000000000000001
Gauss-Piv	2.2204e-16	9.656100799999998e+03	1.0000000000000000 2.0000000000000000 3.0000000000000000 4.0000000000000001

Indice di condizionamento: 4.356292969021429

$$3. A = \begin{pmatrix} 0 & -1 & 2 \\ 2 & -12 & 3 \\ 1 & -2 & 4 \end{pmatrix}, b = \begin{pmatrix} -31 \\ 5 \\ 13 \end{pmatrix}, sol = \begin{pmatrix} 75 \\ \frac{197}{21} \\ -\frac{227}{21} \end{pmatrix};$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive	Elemento nullo sulla diagonale	-	
Gauss-Piv	0	21	75.00000000000000 9.380952380952381 -10.809523809523810

Indice di condizionamento: 29.611023449067236

$$4. A = \begin{pmatrix} 3 & -1 & 2 \\ 2 & 0 & 3 \\ 1 & -2 & 0 \end{pmatrix}, b = \begin{pmatrix} -31 \\ 5 \\ 13 \end{pmatrix}, sol = \begin{pmatrix} -35 \\ -24 \\ 25 \end{pmatrix};$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive	6.0904e-16	7.000000000000003	-34.99999999999986 -23.99999999999979 24.99999999999989
Gauss-Piv	1.0151e-16	7.000000000000001	-35.00000000000000 -24.00000000000000 24.99999999999996

Indice di condizionamento: 8.485209389457932

$$5. A = \begin{pmatrix} 3 & -1 & 2 \\ 2 & -2/3 & 2 \\ 1 & -2 & 1 \end{pmatrix}, b = \begin{pmatrix} -31 \\ 5 \\ 13 \end{pmatrix}, sol = \begin{pmatrix} -\frac{381}{10} \\ -\frac{63}{10} \\ \frac{77}{2} \end{pmatrix};$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive	Elemento nullo sulla diagonale	-	
Gauss-Piv	1.8456e-16	3.333333333333334	-38.09999999999994 -6.300000000000000 38.49999999999993

Indice di condizionamento: 11.475872022165140

$$6. A = \begin{pmatrix} \frac{1}{5} & \frac{3}{5} & \frac{11}{10} & \frac{3}{5} \\ \frac{1}{10} & \frac{16}{5} & \frac{12}{5} & \frac{2}{5} \\ \frac{1}{10} & -\frac{13}{5} & -\frac{13}{10} & 2 \\ \frac{1}{10} & -\frac{13}{5} & -\frac{9}{10} & \frac{2}{5} \end{pmatrix}, b = \begin{pmatrix} \frac{53}{50} \\ \frac{303}{50} \\ -\frac{152}{25} \\ -\frac{126}{25} \end{pmatrix}, sol = \begin{pmatrix} \frac{3}{5} \\ \frac{9}{5} \\ \frac{1}{5} \\ -\frac{3}{5} \end{pmatrix};$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive	5.6659e-01	-0.417600000000000	-0.475862068965512 1.608620689655172 0.500000000000000 -0.600000000000001
Gauss-Piv	9.3989e-15	-0.417600000000000	0.600000000000018 1.800000000000002 0.199999999999996 -0.600000000000001

Indice di condizionamento: 1.214384091227094e+02

$$7. A = \begin{pmatrix} 1 & -4 & 0 \\ 3 & -12 + 10^{-12} & 3 \\ \frac{1}{2} & 1 & 2 \end{pmatrix}, b = \begin{pmatrix} 5 \\ 27 - 10^{-12} \\ \frac{15}{2} \end{pmatrix}, sol = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix};$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive	1.7762e-03	-8.999999999998000	0.992895204262878 -1.001776198934281 4.000000000000001
Gauss-Piv	6.1062e-16	-8.999999999997998	1.000000000000002 -1.000000000000000 4.000000000000000

Indice di condizionamento: 47.984855409493903

$$8. A = \begin{pmatrix} 1 & -4 & 0 \\ 3 & -12 + 10^{-15} & 3 \\ \frac{1}{2} & 1 & 2 \end{pmatrix}, b = \begin{pmatrix} 5 \\ 27 - 10^{-15} \\ \frac{15}{2} \end{pmatrix}, sol = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix};$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive	1	-8.999999999999996	-3.000000000000000 -2.000000000000000 4.000000000000001
Gauss-Piv	5.8287e-16	-8.999999999999995	0.999999999999998 -1.000000000000001 4.000000000000001

Indice di condizionamento: 47.984855409489072

$$9. A = \begin{pmatrix} \frac{1}{5} & \frac{3}{5} & \frac{11}{10} \\ \frac{1}{10} & \frac{16}{5} & \frac{12}{5} \\ \frac{1}{10} & -\frac{13}{5} & -\frac{13}{10} \end{pmatrix}, b = \begin{pmatrix} \frac{12}{25} \\ -\frac{3}{5} \\ \frac{6}{25} \end{pmatrix};$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive		-2.575717417130363e-16	6.783352818053212e + 15 -1.206654107057543e + 15 1.891511843495608e + 15
Gauss-Piv		-2.575717417130363e-16	-6.783352818053212e + 15 -1.206654107057543e + 15 1.891511843495608e + 15

Indice di condizionamento: 4.535198569220834e+16

Rango $A = 2$, Rango $B = 3$.

$$10. A = \begin{pmatrix} 2 & 5 & 4 \\ 0 & 0 & 10 \\ 8 & 2 & 20 \end{pmatrix}, b = \begin{pmatrix} 11 \\ 10 \\ 30 \end{pmatrix}, sol = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix};$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive	Elemento nullo sulla diagonale	-	
Gauss-Piv	0	360	1 1 1

Indice di condizionamento: 7.882998953945470

$$11. A = \begin{pmatrix} 3.3 & 4.1 & 1 \\ 0.3 & 4 & -1 \\ 3 & 0.1 & 2 \end{pmatrix}, b = \begin{pmatrix} 8.4 \\ 3.3 \\ 5.1 \end{pmatrix};$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive	-	2.657873920952625e-15	8.443609022556391 -2.308270676691729 -10.000000000000000
Gauss-Piv	-	2.657873920952625e-15	8.443609022556391 -2.308270676691729 -10.000000000000000

Indice di condizionamento: 4.227405868685302e+16

Rango $A = 2$, Rango $B = 2$.

$$12. \ A = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \end{pmatrix}, b = \begin{pmatrix} \frac{25}{12} \\ \frac{77}{60} \\ \frac{19}{20} \\ \frac{319}{420} \end{pmatrix}, sol = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Metodo	Errore relativo	Determinate	x
Gauss-Naive	1.3145e-13	1.653439153439211e-07	0.999999999999994 1.0000000000000062 0.9999999999999869 1.0000000000000078
Gauss-Piv	1.4522e-13	-1.653439153439300e-07	0.999999999999993 1.0000000000000069 0.9999999999999855 1.0000000000000085

Indice di condizionamento: 1.551373873892904e+04